Fiscal Year:	FY 2009	**Task Last Updated:**	FY 02/16/2009
PI Name:	Paloski, William H Ph.D.		
Project Title:	Spatial Reorientation of Sensorimotor Balance Control in Altered Gravity		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Behavior and performance		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Sensorimotor**:Risk of Altered Sensorimotor/Vestibular Function Impacting Critical Mission Tasks		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	whpaloski@uh.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	(713) 743-9272
Organization Name:	University of Houston		
PI Address 1:	Health & Human Performance		
PI Address 2:	3855 Holman St, Garrison 104		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77204-6015	**Congressional District:**	18
Comments:	PI moved to University of Houston in September 2008, from NASA JSC.		
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-02
Start Date:	10/01/1999	**End Date:**	01/13/2009
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	McCollum, Suzanne	**Contact Phone:**	281 483-7307
Contact Email:	suzanne.g.mccollum@nasa.gov		
Flight Program:	Shuttle		
Flight Assignment:	NOTE: HRP Master Task List dated 1/11/2012 shows end date of 1/13/2009 (Ed., 2/23/2012) NOTE: Project extended until 05/31/2008 per PI (3/08) NOTE: Project extended until 03/31/2008 per PI (5/07) Pre-/Postflight Only In flight development phase (data collection has begun)		
Key Personnel Changes/Previous PI:	N/A		
COI Name (Institution):	Wood, Scott Ph.D. (Universities Space Research Association) Kaufman, Galen Ph.D. (University of Texas Medical Branch) Reschke, Millard Ph.D. (NASA Johnson Space Center) Black, Franklin M.D. (Legacy Health Sysytem)		
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Disruption of balance control is one of the most significant post-flight manifestations of sensorimotor adaptation to microgravity. This investigation examines changes in spatial processing of sensorimotor function following adaptation to microgravity, and explores the feasibility of forcing changes between motor program sets optimized for different gravitoinertial environments. This study involves pre- and post-flight measurements on eleven astronaut subjects (first-time fliers) following short-duration Shuttle missions (11-13 days). Each astronaut participates in one familiarization training session (including standard posturography testing), three pre-flight data sessions, and four post-flight data sessions on R+0, 2, 3, and 7/8 days. Each test session includes posture measurements, and three sessions (L-60, L-30 and R+3) include additional posture measurements immediately after exposure to short-radius centrifugation. In addition, testing on normative subjects (ongoing) support data interpretation and establish ground-based controls. Our first specific aim is to examine adaptive changes in the spatial reference frame used for coding orientation and motion as a function of space flight. By examining the effects of head tilt on balance control, we will test our hypothesis that there is a reorientation of central vestibular processing from a gravitational frame of reference to a head frame of reference as a function of adaptation to microgravity. Our second specific aim is to examine the feasibility of altering the re-adaptation process following space flight by providing discordant canal-otolith-somatosensory stimuli using short-radius pitch centrifugation. This study specifically addresses Neurovestibular Adaptation questions of the Critical Path Roadmap (CPR Risks 15 and 16) related to the physiological basis for disruption of balance control and spatial disorientation following g-transitions. Additional post-flight posturography measures obtained during this study will expand our existing database and increase our ability to predict recovery curves from mathematical models for individuals based on pre-flight postural performance. The incorporation of head tilts during posturography addresses our general hypothesis for the physiological basis for the disruption of balance control following g-transitions, and provides an enhanced technique to determine crew readiness to return to normal activities following return to Earth. The incorporation of short-radius centrifugation during the post-flight period addresses the ability of crewmembers to adapt to multiple gravitoinertial environments and switch between them, e.g., as required for intermittent exposure to artificial gravity. The results of this study will also assess the potential risk for disrupting the re-adaptation process by exposure to discordant sensory stimuli during low levels of centrifugation.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study has demonstrated the increased sensitivity of computerized dynamic posturography (CDP) with the incorporation of static and dynamic head movements and a sway-referenced platform. This enhanced CDP protocol seems more suitable for monitoring astronaut fitness for return to duty, and may be useful in the early detection of balance disorders.
Task Progress:	During this final reporting period, the analysis phase was completed. During this period we utilized Receiver Operator Characteristic (ROC) curves to provide a means of concurrently evaluating each test condition's sensitivity and specificity. Since the standard clinical neurological assessments are typically constrained to firm support surfaces and head-erect postural orientations, the primary objectives of these analyses were to contrast the diagnostic performance between CDP tests conducted using fixed and unstable support surfaces and between CDP tests conducted using head erect and dynamic head-tilt conditions. Sensitivity is defined as the true-positive rate, i.e., the proportion of correctly classified cases with a given condition. Specificity is defined as the true-negative rate, i.e., the proportion of cases without the condition correctly identified. Sensitivity and specificity trade off against one another as the classification threshold varies. ROC analysis can be used to assess a test's performance by comparing its sensitivity with specificity over a range of thresholds. The ROC curve is a plot of sensitivity versus 1-specificity as the classification threshold changes. In particular the area under the ROC is a useful index for evaluating the inputs to the classifier. In our case, we evaluated the individual test conditions used to distinguish astronauts from controls on landing day in terms of their respective ROC areas. A rough guide to interpreting the area under the curve is the traditional academic point system: 0.90-1.0 = excellent; 0.80-0.90 = good; 0.70-0.80 = fair; 0.60-0.70 = poor; 0.50-0.60 = fail. Sensitivities and specificities were used to determine optimal thresholds for assessing postural stability recovery. The area under the ROC curve was 0.77 (fair) for the standard eyes-closed fixed-support Romberg test, improving to 0.86 (good) with head movements. With the unstable support, however, post-flight performance was so degraded that for a range of thresholds, astronauts could be completely distinguished from controls, with or without head movements (ROC area = 1.0). In situations, where the ROC area is less than 1.0, it is desirable to choose a threshold that gives high sensitivity at the expense of lower specificity so that astronauts who have not recovered will almost certainly be identified. For example, when using the eyes-closed fixed-support Romberg test with head movements, one might use the threshold that gives a sensitivity of 0.91 and a specificity of 0.64. By comparison, increased specificity (91% of recovered astronauts, identified), but with sensitivity dropping to 0.82 (18% of non-recovered astronauts erroneously classified as recovered). A more conservative threshold can be used to correctly classify all astronauts at 100% test sensitivity, but with only 18% specificity (orange circled point). Thus using the ROC curve, one can make a ready evaluation of test sensitivity and specificity providing an evidence-base for assessing post-flight recovery. ROC areas for the classifiers were derived from the same data used to estimate them (only 22 subjects). As a result, when applying the classifier to EQ scores from future missions the actual ROC areas would be expected to be somewhat lower. To obtain a more realistic prediction of the sensitivity and specificity for this classifier, Bayesian analysis was used to estimate the statistical model for EQ scores. This model was then used to simulate data that could be plausibly obtained from 1000 additional subjects. Values of predicted sensitivity and specificity obtained by classifying the simulated data were 0.943 and 0.973 (as opposed to both being 1.0 as estimated from the original data). For the unstable platform without head movements, even though the ROC area was 1.0 for the sample of 22 subjects, the simulation-predicted sensitivity and specificity were 0.918 and 0.734, both considerably lower than for the analogous test with head movements.
Bibliography Type:	Description: (Last Updated: 02/09/2021)

Abstracts for Journals and Proceedings	Feiveson AH, Wood SJ, Jain V, Paloski WH. "Using an informative missing data model to predict the ability to assess recovery of balance control after spaceflight" Presented at the American Statistical Association Joint Statistical Meetings, Denver CO, August 5, 2008. American Statistical Association Conference Proceedings, August 2008. , Aug-2008
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Articles in Other Journals or Periodicals	Wood SJ, Black FO, Paloski WH. "Destabilization of human balance control by head tilts following space flight." Journal of Neurophysiology, (in preparation), February 2009. , Feb-2009
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